

MID-TERM EXAMINATION - I

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

First Semester, September - October, 2025

Paper Code: ETCH-103**Subject: Applied Chemistry****Time: 1½ Hrs.****Max. Marks: 30****NOTE: Attempt three questions. Question No 1 is compulsory. All questions carry equal marks.**

Q. No.	Question	Marks	CO(s)
1.a	Why and how are corrections made in determination of calorific value by bomb calorimeter?	2	CO1
1.b	Hydrocarbons that are poor gasoline fuels are quite good diesel fuels. Explain with chemical reasoning.	2	CO1
1.c	What is an invariant system? Explain with a suitable phase rule example.	2	CO2
1.d	In the phase diagram of ice, the fusion curve of ice has a negative slope. Explain using Clapeyron equation.	2	CO2
1.e	What are the advantages of catalytic cracking over thermal cracking in petroleum refining?	2	CO1
2.a	What is carbonization? Explain with neat diagram, the Otto-Hoffman by-product oven method for the manufacturing of metallurgical coke.	6	CO1
2.b	A coal sample has following composition by weight: C = 90%, H = 6%, O = 3%, S = 0.5%, N = 0.5% . Calculate Net and Gross calorific value using Dulong's formula. (Latent heat of condensation of steam = 587 cal/g).	4	CO1
3.a	Draw a neat, labelled phase diagram of Sulphur system and explain areas, curves, triple point and metastable state.	6	CO2
3.b	State Phase Rule. Discuss its application to one component Water system with a neat phase diagram.	4	CO2
4.a	Write short notes on: (i) Octane number and Cetane number, (ii) Power alcohol.	6	CO1
4.b	Explain Proximate and Ultimate analysis of coal along with their significance.	4	CO1

****** End of Question Paper ******